

Impact of Introducing Technical Design Elements in Makerspace Trainings

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This work provides an excerpt of research done at MIT towards my Master's thesis of the same title [1].

Abstract

Makerspaces are used as a tool in higher education to support curricular, hands-on projects and encourage student extracurricular and personal projects. Because access to making is more self-driven, there is a gap between what makerspace trainings teach students and what students are expected to know by the time they reach capstone courses in engineering. To test the effects of introducing a technical makerspace training to students, several steps were taken. First, Design Expertise was defined to combat barriers to makerspace use: it is a combination of knowledge, skill, perspective, and motivation. Second, a rigorous framework, the Design-Fabrication-Performance (DFP) matrix was created to break down design expertise into manageable chunks. Next, existing makerspace trainings at MIT were characterized using the DFP matrix. Afterwards, the DFP matrix was used to design a new experimental training which would incorporate engineering design thinking and expertise with the typical makerspace machine training structure. Finally, 23 student participants were recruited, trained, surveyed using a Likert scale (1 = strongly disagree, 5 = strongly agree), and interviewed to understand the impact of the training on participant perspectives, engineering identity, and maker motivation.

The maker training reinforced the motivation to make things for most students, with the average score for the associated question being 4.48 (SD: 0.85). The training also positively impacted some traditionally marginalized groups in STEM. For the statement "I feel comfortable in engineering at MIT", women averaged 3.27 and men 3.90 before the training. The average differentials of the post- and pre-training scores for this question for these groups were 0.91 and 0.4 respectively. The training also appeared to level the playing field for students with less advanced STEM backgrounds. For the question "I am confident in my ability to solve GIR level problems on my own", students with parents with graduate degrees or higher averaged 4.44 before the training, while those with parents with undergraduate degrees or lower averaged 3.57. The average differentials are 0.22 and 0.64 respectively. A longitudinal study following these students would be needed to reveal long term outcomes such as STEM retention and long-term makerspace usage.

Introduction and Related Work

Makerspaces in the realm of higher education have emerged as a tool for students to expand on theoretical STEM learning through hands-on academic, extracurricular, and personal projects [2], [3]. Few papers have delved into applying pedagogical methods or documenting specific projects that incorporate making into the undergraduate engineering curriculum [4], [5], [6]. Current gaps in the literature do not address advanced making needs or initiatives, or describe projects created using advanced tools or engineering theory. In terms of maker pedagogy and engineering education, project-based capstone classes are traditionally introduced late into a student's degree program [2], [7], so access to making is more self-driven early on. As students can select the specific courses they want to take to complete their degree requirements, gaps in student experience and expertise naturally arise, with some students never having used a makerspace before their hands-on capstone classes. On top of that, there is a worry that makerspaces will exacerbate inequalities found in STEM, so there is preliminary research exploring ways to diversify makerspaces and build a welcoming culture [8], [9], [10], but these papers do not suggest or test any new policies which may result in improvements in makerspace use.

To democratize access to advanced making and rigorous engineering design, introducing access to design expertise to undergraduate students is proposed. Design expertise is defined as a combination of knowledge, skill, perspective, and motivation. It is hypothesized that having design expertise access increases student self-efficacy and confidence in STEM and retains students in STEM fields.

The *intent* of this research is to understand student motivations and barriers to making, key design expertise factors, optimal methods to present knowledge and skills students need to learn, makerspace and class needs assessment, and design expertise deployment. The proposed design expertise is *important* as it can increase makerspace use, STEM retention, and the number of successful projects, leading to a feedback loop of better outcomes and increased funding. This can significantly benefit students (increased maker confidence), faculty (pedagogy improvements, attract more funding), makerspace staff (support complex maker projects, attract funding/resources), and stakeholders (positive investment outcomes). Creating and deploying

design expertise will positively *impact* several factors. Student making experiences should improve, as shown by increased self-efficacy and positive feelings towards makerspaces and makerspace community. The number of student prototypes, projects, and engineering designs undertaken will grow not only in quantity, but also complexity, dynamism, and fidelity. Finally, the number of students who continue in STEM should also increase: increased enrollment in STEM majors and minors, more frequent use of makerspaces for academic and personal projects, and positive industry and economic outcomes.

Design Expertise

Access to design expertise is meant to address barriers to entry and democratize access to STEM concepts and maker tools. Design expertise as proposed in a conversation with Martin Culpepper is defined as a combination of (1) knowledge, (2) skill, (3) perspective, and (4) motivation. Knowledge is defined as engineering content and STEM concepts, made accessible and clear for students. Skill refers to the capability of using maker tools and machines to create dynamic projects. Perspective is defined as best practices in engineering design and making, and understanding how knowledge and skill fit together optimally. Motivation relates to the ways students take initiative towards making, which can be influenced by case studies, personal interest, and encouragement.

Fig. 1 shows the four components of design expertise and how they relate to each other. It is proposed that students need all four components to be able to make things in a manner that is thoughtful, efficient, and rigorous.

Most students likely identify with one of the sections labeled A-J. “Intuitive” makers refer to students who have experience using machines and a certain intuition for using them but lack the knowledge of STEM concepts and motivation to learn them. “Theoretical” makers refer to students who have taken STEM classes and perhaps even observed maker content but are missing the hands-on skills and motivation to get into the

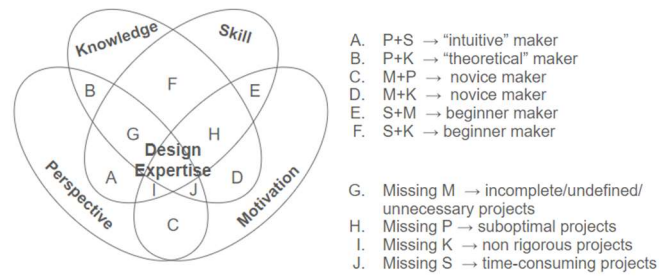


Figure 1: Design expertise framework presented as a four ellipse Venn Diagram.

makerspace and build things. Novice makers are students who have the motivation to make and learn, paired with either perspective or knowledge, but are missing out on hands-on maker skills. One step above that are beginner makers, students who have some hands-on making skills paired with either motivation or knowledge but lack the perspective and best practices. The design expertise framework, a combination of all four components, is proposed to give students accessible tools, guidance and confidence to make complex projects and use advanced machines. As a result, we hypothesize that access to design expertise will increase overall participation, student self-efficacy, and diversity in makerspaces.

Design-Fabrication-Performance (DFP) Matrix

To break down design expertise into sub-components, a Design-Fabrication-Performance (DFP) matrix is proposed, developed through interviews with makerspace experts, staff, student mentors, and users; it presents design processes and outputs as categories on the X and Y axes respectively. These are arranged in order of complexity. Design processes proposed include benchtop machines, 3D printers, laser cutters, waterjets, mills and lathes. Along the same axis are assembly processes as well, where assembly splits into alignment and fixation. Alignment includes threads, pins/slot/hole, and contacts, which refer to any type of mating surface or fit features. Fixation includes

		Processes					Assembly					
							Alignment			Fixation		
		Benchtop Machine	3D Print	Laser	Waterjet	Mill	Lathe	Threads	Pins/ Slot/ Hole	Contacts	Adhesives	Fasteners
Output	Static Interfaces/ Features											
	Shafts											
	Structural Elements											
	Bearings											
	Gears											

Figure 1: Overall Design-Fabrication-Performance (DFP) matrix. The shaded cells are a characterization of a first-year maker training at MIT (“Digital Fabrication” at Metropolis) using the DFP matrix.

adhesives, fasteners, which are inclusive of rivets, bolts, screws, and more, and friction, which includes interference fits and snap fits. Design outputs on the Y axis include static interfaces/features, shafts, structural elements, bearings, and gears. Fig. 2 shows the whole DFP matrix.

There are several ways this matrix can be used. The first is as a learning tool. In each cell of the matrix, students will be presented with a breakdown of the knowledge, skills, best practices, and motivation needed to achieve proficiency in designing the output for a given maker tool as input. Working with different materials on different machines leads to certain considerations to keep in mind.

A second way the DFP matrix can also be used is to characterize projects and the different methods and machines used to create them. This can either help students select projects to make which are appropriate for their experience or help students find and fill gaps in their own maker journey.

A third way of using the DFP matrix is to characterize maker trainings. This can reveal gaps in student learning and also be a means for makerspace mentors to design new workshops based on specific desired learning outcomes. In this way, student learning is also able to be characterized in a manner that is more visual and intuitive compared to survey metrics or other theoretical measurement constructs [11], [12], [13]. Further metrics can be developed to enhance the amount of data coded by the matrix. An example is shown in Fig 2, where a beginner training at MIT is characterized using the matrix. If the matrix colors are set to be translucent, matrices from different trainings can be superimposed onto each other for students to see how they developed their expertise over time and how their knowledge, skills, and perspective have grown. It can also help students figure out the gaps they have left to fill.

The last way the DFP matrix can be used is to find gaps in existing trainings and classes and create new trainings to fill them. By knowing which cells in the matrix to target, makerspace staff and educators can create specific interventions for their users.

Training Design

The DFP matrices for different trainings at MIT (such as the one shown in Fig. 2) reveal a gap between what students learn and what they are expected to know as upperclassmen. Most trainings focus more on how to use particular machines. While a beginner laser cutter training may provide experience using software to personalize a static object like a keychain, engineering classes often call for students to design and fabricate dynamic, complex components. To remedy this, the DFP matrix was used to optimize the design of a new experimental training that would go beyond traditional and static demo parts to give students as many skills as possible within a given time frame, addressing as many makerspace barriers as possible.

For students, the training should have a high “bang for the buck”, delivering maximum useful information in minimal time. The training environment should be welcoming so students can feel comfortable asking questions and testing

new ideas regardless of whether they “succeed” or “fail”. The training should emphasize how math, science, and modeling tools feed into design best practices of a given object, skills which should be transferable to future projects. The chosen project should be simple enough for first-year undergraduates to complete but appropriately challenging such that student decisions affect the final outcome in tangible ways. The cost per student for materials should be reasonable to be able to train as many students as possible for detailed analysis of outcomes. After requirements such as these were generated, a list of potential gadgets was created and their merits were compared against these criteria. The biggest concern amongst the options explored was introducing variability that could not be controlled. There was also concern that having the project presented as a “competition” may make students feel stressed or compare themselves to one another, which could detract from the goal of fostering a welcoming training environment.

With all of these considerations in mind, the final project chosen was utilizing a car and linear spring launcher with the goal of having the car travel as far as possible. This would be introduced as a design objective, with there being no minimum distance. The primary aim would be to compare the modeled distance to the actual distance traveled and understand how and why any deviations may occur.

Designing the Car

The car was designed in OnShape since it could be accessed by students without any time-consuming software downloads. In terms of the car shape, a streamlined body (“teardrop”) was chosen for its low drag coefficient. The students would have the opportunity to prove mathematically that the teardrop is the best choice for their car. The mass of the car would be a variable the students could choose through picking an infill value in their 3D printing slicer that matched their calculations. The bearings constraining the axle with respect to the car would be pre-chosen, shown in Fig. 3, but the students would walk through a design exercise where they conclude that the bushings chosen achieve the design requirements (including being low cost, ~\$8 for all 8 bushings) and minimize friction torque.

There would be several options for the wheels of the car the students could choose from: 3D printed, laser cut, and waterjet. Rubber bands would also be available if students wanted to increase wheel friction. The axles would also be provided as 3D printed components, but the students would work through a guided design problem to help them converge on the design provided to them, shown in Fig. 4. This design exercise would highlight the number of different possible solutions and why the particular one was chosen.

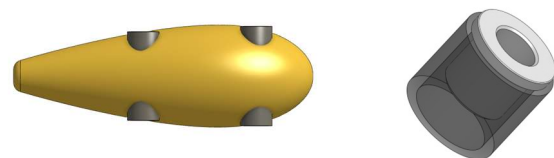


Figure 32: (Left) bottom view of bushings press fit into car body. (Right) views of inner flanged bushing for axle (white) and outer bushing for car (gray). A small gap between them is needed to minimize friction, approximately 1/16”.



Figure 4: CAD isometric view of final car model and CAD section view of two-part axle (blue), connector piece (pink), flanged bushing (white), and wheels (black). Left axle shoulder is longer than right axle shoulder due to the teardrop shape of the car.

Designing The Launcher

To design the “launcher”, several energy storage methods were considered, including gravitational potential energy (GPE), spring energy, and beam bending energy. Spring energy was the chosen as the equations governing springs in the linear regime are simple and contain few variables:

$$F_{spring} = kx \quad (1)$$

$$PE_{spring} = \frac{1}{2}kx^2 \quad (2)$$

With springs, the main two determinants of energy stored are the spring constant, k , and the displacement, x . The maximum displacement x for any given k would be linked to the maximum force with which a given student is able to load the spring. The strength of students varies, but it was estimated to range between 30 and 70 Newtons. Since displacement is squared in the equation for potential energy (2), it is advantageous for students to choose a spring with a lower spring constant and higher displacement. The final two springs chosen for the training approximately had spring constants of 1.6 N/mm and 5.4 N/mm, with other springs present as demo pieces. These specific spring options were chosen to give students an explicit choice, for their low costs, and easy availability on Amazon.

The spring launcher was designed so that students would not have to engage in any CAD modeling themselves. This would allow them to focus on modeling the system and choosing a spring and displacement desired. The launcher was

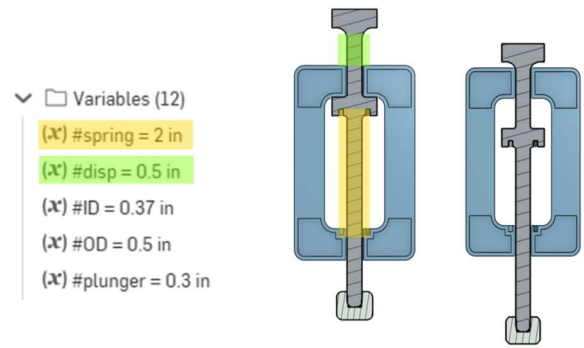


Figure 5: Parametric CAD variables. Highlighted variables are student choice. A fully engaged launcher is shown on the right. Plunger (gray) is one piece, with end cap to secure spring.

created in OnShape parametrically so that students could change individual variables to update the CAD model, shown in Fig. 5. Once the “plunger” was pulled back all the way, a hard stop would be hit to indicate that the desired displacement was reached, storing a known and modeled amount of energy. The final DFP matrix for the new training is shown in Fig. 6.

Designing the Worksheets

After the car was designed, worksheets were created highlighting the relevant equations and modeling of different components of the car. A copy of the worksheets can be found in Appendix A and a summary of the content is provided as follows. The first worksheet was intended to be a review of 8.01 (Classical Mechanics) physics content, including force body diagrams, work, and energy. The focus was not to obtain an exact answer but rather to understand magnitudes of different units to help students build their engineering intuition. The second and third worksheets covered content relating to the two resistive forces the car would face and that the students would focus on modeling: drag force and bearing friction torque. Spreadsheet modeling was used to explore how drag coefficients are calculated. The bearing friction worksheet focused on a more conceptual understanding of the

		Processes						Assembly					
								Alignment			Fixation		
		Benchtop Machine	3D Print	Laser	Waterjet	Mill	Lathe	Threads	Pins/ Slot/ Hole	Contacts	Adhesives	Fasteners	Friction
Output	Static Interfaces/ Features												
	Shafts												
	Structural Elements												
	Bearings												
	Gears												

Figure 6: Final DFP matrix characterization for the car training.

	A	B	C	D
1	Setting Initial Conditions for Energy			
2	Variables	Value	Units	Description
3	YOUR INPUT			
4	m	100	g	Mass of car + wheels + axle
5	m	0.1	kg	Mass of car + wheels + axle
6	k	2	N/mm	Spring constant, experimental
7	k	2000	N/m	Spring constant converted
8	x	1	in	Displacement in in
9	x	0.0254	m	Convert displacement to m
10				
11	OUTPUTS			
12	PE	0.64516	J	Energy stored in spring, $1/2 \cdot k \cdot x^2$
13	F	50.8	N	Force required to load spring
14	v_i	3.592102448	m/s	Initial velocity

Figure 7: Initial conditions for energy spreadsheet, supplementing worksheet 4.

differences between bearings and bushings and why students would choose one over the other. The fourth worksheet focused on energy modeling in three sections: spring potential energy, the overall energy modeling of the car including resistive forces, and creating an equation of motion (EOM) representing the dynamics of the car. The EOM was programmed into a MATLAB live script which students could use to visualize plots of velocity and distance to further optimize the mass of their vehicle. Fig. 7 shows an example of the spreadsheet used to help students optimize their initial velocity. It was organized so that inputs in yellow cells would cause the red text to auto-calculate.

Survey and Interview Design

A survey was conducted twice, before and after the students completed their training, to assess how their attitudes changed as a result. While most questions in the pre- and post-training surveys overlapped to enable direct comparison of responses, there were a few differences. The pre-survey incorporated demographics questions including gender, race, and the highest level of parent education, among others. The post-survey included training specific questions to assess the quality of the training design and achievement of learning outcomes. The questions that remained the same in both pre- and post-surveys plus training specific questions in the post-survey were presented with a Likert scale from 1 to 5, with 1 being strongly disagree and 5 being strongly agree. These questions were adapted from an existing survey created by Rima Das, Amanda Baker, and others on the NSF grant research team. Questions can be found in Appendix B. There are limitations to using survey data, as they can introduce biases; hence a follow-up interview was also designed to provide a deeper understanding of student perspectives.

Sample Selection

Screening survey results were filtered for students who previously took at least a beginner makerspace training. Students also had to have taken 8.01 Physics I or equivalent. Students had rated their experience with different makerspace machines from 1 to 5 (no experience to confident, independent use). Most students selected had an average rating of around 2, indicating that while they may have had training, they had little experience beyond it. This selection of

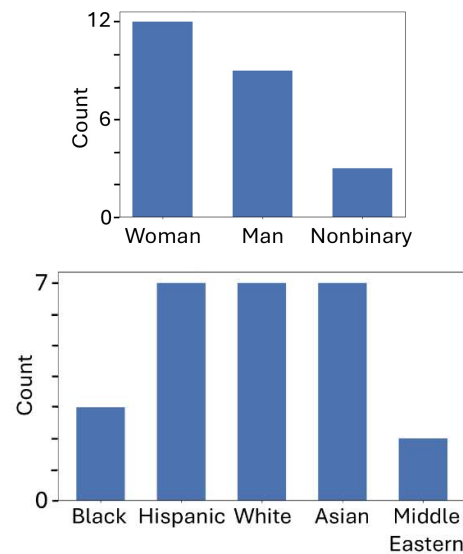


Figure 8: Gender (above) and race (below) distribution of sample. N=23.

students is particularly interesting to explore, as several different reasons could explain why they did not progress in their experience with makerspace machines. The last step was generating the final sample based on availability and demographics, primarily gender and race. A final group of N=23 students was selected. Fig. 8 shows the plots of gender and race distribution of the final 23 students who were selected and completed the study.

Observations During Training

Overall, going through the worksheets with the students went smoothly. The content appeared to be too easy for some, while at an appropriate level for others who were revisiting it after some time. Although the variation in difficulty is to be expected, additional questions or activities for those who were waiting for others to finish could have proven to be useful. Although most students had no formal training in solving differential equations, they appeared to understand the process to obtain the system's equation of motion. The MATLAB script was introduced towards the end of the training, with mixed results. Some appeared to understand how to use the code, while others felt unsure. Most students indicated that they had never used MATLAB, but all students mentioned some previous level of coding experience.

Besides the students who needed extra parts because they did not get the chance to 3D print, some students had several issues while 3D printing or post-processing parts which meant they needed replacements. All of these issues were unexpected; despite this, these "failure modes" often provided good lessons for the students, and having spare parts meant students did not necessarily lose out on having a finished car they could launch. The students overall displayed a lot of interest in trying to understand why their cars did or did not behave as intended. Once again, despite these "failure modes", students displayed high levels of resilience and problem-solving skills. Most students responded to these situations by diagnosing the problem and experimenting with different methods to resolve it.



Figure 9: Students launching their cars. Some students used their hands to hold down the launcher, while others used their feet.

Results and Discussion

Much of the literature on diversity in makerspaces highlights the lack of belonging that women feel. Analyzing this split of the data was critical in understanding potential ways to mitigate the gender gap. Because the sample size of nonbinary students was small, the data was split into 14 women and 7 men. Fig. 10 relays histograms for survey question 10: **“I feel comfortable in engineering at MIT”**. The average pre-training scores for men and women are notably lower, with the distribution for men centered around “slightly agree” and the distribution for women centered closer to “neither agree nor disagree”. The distributions more visibly shift after the training, particularly for women, whose average increases by almost an entire point. The gap between women and men’s sense of comfort in engineering at MIT decreases in size post-training, equalizing the playing field. This could possibly indicate that the training experience provided some amount of reassurance or comfort in women’s engineering identity, either through the training content itself, the hands-on building experience, or even relating to the identity of the workshop facilitator.

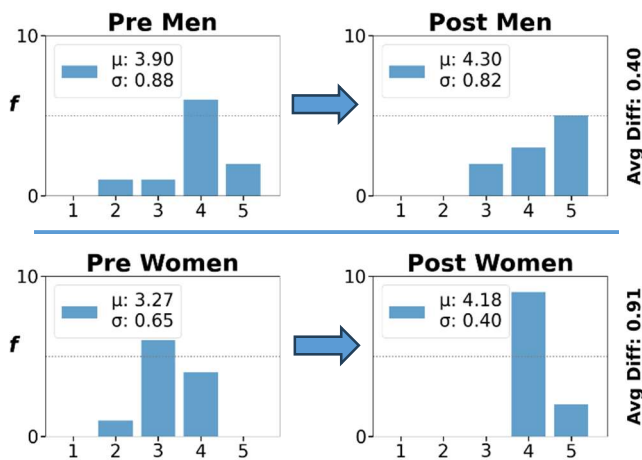


Figure 10: Pre and post survey responses for men and women plotted for Q10: “I feel comfortable in engineering at MIT”.

Fig. 11 shows histograms for question 14: **“I am confident in my ability to solve General Institute Requirement (GIR) level problems on my own”**. The gap between men and women is sizable, with the pre-training distributions offset from one another by almost a whole point: this coincides with previously reported lower STEM self-efficacy in women. There is once again an “equalizing” effect post-training, but less pronounced. The women’s average

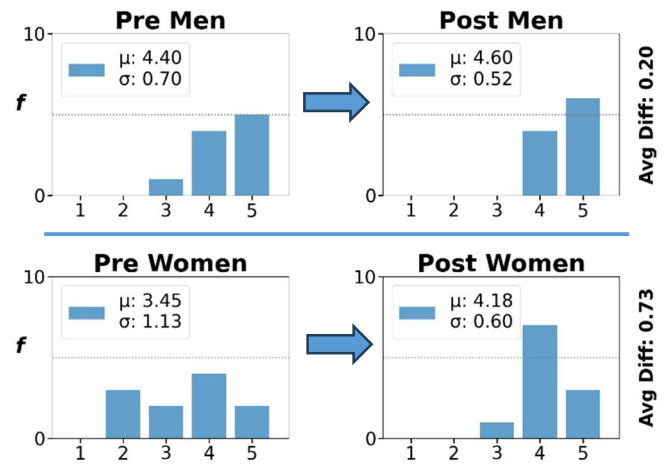


Figure 11: Pre and post survey responses for men and women for Q14: “I am confident in my ability to solve GIR level problems on my own”.

differential is nearly four times higher than the men average differential, but a gap does persist

The effect of parents’ highest level of education may be linked to the volume of a student’s STEM experiences or higher levels of self-efficacy before going to college. Students with highest parent education of undergrad degree and below had a sample size of 14, and grad degree and above had a sample size of 9. Fig. 12 shows histograms for question 14: **“I am confident in my ability to solve GIR level problems on my own”**. Students in the undergrad-lower group had lower means than the students in the grad-higher group, with almost a whole point difference for question 14. Visually, the distributions reflect this disparity. Post-training, the students in undergrad-lower catch up and the grad-higher group experiences less growth. This may indicate that those with parents who have a grad degree or above have had more opportunities to engage with higher-level STEM material before coming to college compared to students with parents who have an undergrad degree or below. The training experience appears to level the playing field a little more.

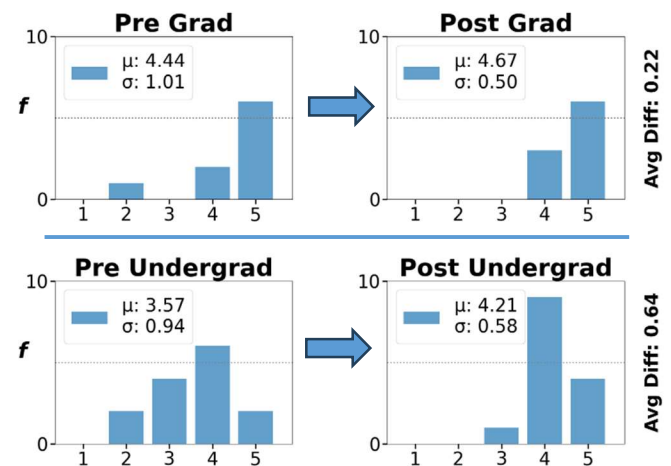


Figure 12: Pre and post survey responses for highest parent education plotted for Q14: “I am confident in my ability to solve GIR level problems on my own”.

Interview Insights

The interviews revealed key pieces of information not revealed in the survey. Less than half of the students mentioned that they liked the training specifically because it integrated modeling with making. The favorite parts of the training cited most often were the assembly of the car and testing/launching the car, while the least was doing the calculations, mostly the physics review. More work can be done in the future to make technical and conceptual discussions more engaging. Overall, students affirmed they generally had more confidence in choosing maker processes for specific design requirements, with slightly less reporting confidence in their ability to design their own project using “math and engineering”. Most students mentioned that this training did not change their beliefs about makerspaces, but rather that it reinforced their positive beliefs. About a quarter of the students said their beliefs about makerspaces had changed, noting that such spaces now felt more accessible and welcoming. A little more than half of the students said the training made them more confident and comfortable in their own engineering ability (self-efficacy), highlighting how while they may have already possessed the necessary knowledge, they now felt more ready to utilize it.

Conclusion

This paper has demonstrated that introducing technical aspects into makerspace trainings can achieve positive outcomes, particularly increased student self-efficacy and feelings of belonging to the makerspace, as well as motivation to continue making. These technical aspects were introduced as part of a broader effort to characterize design expertise in students, so makers and educators can track their progress and fill gaps in their knowledge. The training was designed using the DFP matrix to try and cover skills not taught in current maker trainings at MIT. Several of the distribution plots aligned with the literature citing lower self-efficacy values for women compared to men. For some questions, increases in self-efficacy were equal for men and women, but since women started at lower averages, gaps in scores persisted. There is a need to further understand why gaps persist in some survey metrics and disappear in others.

Key contributions to the study of makerspaces and makerspace pedagogy are (1) developing the DFP matrix to break design expertise into manageable blocks. This is a new way to characterize makerspace trainings, hands-on course content, and student engineering design progress. Another is (2), designing, documenting, and disseminating a technical training so other spaces can adopt or adapt it.

There are many avenues to explore for future work that may improve results, expand student experience, or otherwise provide different insights. Several limitations have already been discussed, but the issues with having a smaller sample size cannot be overstated. If this experiment were to be repeated with more students, different kinds of statistical analysis could have become available which would have made results clearer and likely more significant. It would be helpful for similar technical trainings to be adopted in other academic makerspaces, to draw conclusions about training

efficacy in varied contexts. More trainings can be characterized using the DFP matrix to find gaps, and more new trainings can be developed using the matrix. The DFP matrix framework can be further developed by adding more content and by deploying it in other makerspaces. Overall, this training was a first step in integrating engineering and modeling concepts with maker skills and serves as the beginning of the various steps professors, educators, and mentors can take to help students increase their confidence and competence in STEM.

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Appendix A

https://drive.google.com/file/d/1D_pfKQCyfAXGvPWaTav07WHyT25EO5H3/view?usp=sharing

The link above provides the content kit used in the Part 1 training. The first worksheet was intended to be a review of 8.01 (Classical Mechanics) physics content, including force body diagrams, work, and energy. The second and third worksheets covered content relating to the two resistive forces the car would face and that the students would focus on modeling: drag force and bearing friction torque. The fourth worksheet focused on energy modeling in three sections: spring potential energy, the overall energy modeling of the car including resistive forces and creating an equation of motion (EOM) representing the dynamics of the car. Finally, a list of takeaways were provided.

Appendix B

Primary survey questions separated into respective categories. These were compared pre- and post- training using a Likert scale from 1-5.

(1) Makerspace Questions	(2) Engineering Identity
1. Within the next year, I plan on engaging in makerspaces outside of my classes 2. Engaging in makerspaces sounds enjoyable to me 3. Engaging in makerspaces sounds useful to me 4. I know how to find one or more makerspaces on campus 5. I would feel comfortable walking into a makerspace on campus	6. I enjoy learning about engineering 7. I see myself as an engineer 8. I have had experiences in which others recognized me as an engineer 9. I belong in engineering at MIT 10. I feel comfortable in engineering at MIT 11. I am similar to the types of people who pursue engineering at MIT 12. I get along with people in engineering at MIT
(3) Engineering Self-Efficacy	(4) Engineering Motivation
13. I am confident in my ability to do well in GIR level classes 14. I am confident in my ability to solve GIR level problems on my own 15. I am confident in my ability to use GIR level knowledge to design and make things that perform as intended 16. I can use physical tools to build things 17. I can use physical tools to fix things 18. I can work with machines to make things 19. I can assemble and disassemble things. 20. I can use software or digital tools to make things 21. I can design new things. 22. I can recognize changes needed for a design to work	23. I feel motivated to get good grades in my engineering-related classes 24. I feel motivated to learn in my engineering-related classes 25. I feel motivated to use makerspaces for classes 26. I feel motivated to use makerspaces for extracurricular activities 27. I feel motivated to use makerspaces for personal projects 28. I like to build things with my hands 29. I like to lay out a plan based on math and physics before I build things 30. I like to design, build, and iterate if things go wrong instead planning out my designs